

NANSANA MUNICIPALITY MATHEMATICS SCHEME OF WORK FOR PRIMARY SEVEN TERM ONE 2026

TOPIC 1: SET CONCEPTS

LEARNING OUTCOME: The learner demonstrates the knowledge of sets to solve problems in real life situations.

TOPIC 2: WHOLE NUMBERS

LEARNING OUTCOME: The learner manipulates and converts whole numbers.

TOPIC 3: OPERATIONS ON WHOLE NUMBERS

LEARNING OUTCOME: The learner applies the basic concepts and operations achieved to solve real life problems.

TOPIC 4: PATTERNS AND SEQUENCES

LEARNING OUTCOME: The learner relates and applies simple computational skills in real life situations.

P7 TERM I 2026TOPICAL BREAKDOWN

THEME: SETS TOPIC 1: Set Concepts

Subtopics

- 1. Revision on types of sets
- 2. Identifying finite sets
- 3. Identifying infinite sets
- 4. Forming and listing subsets
- 5. Deriving formula for finding the number of subsets and finding number
- 6. Forming and listing proper subsets
- 7. Finding number of proper subsets using the formula
- 8. Determining the number of elements given the number of subsets
- 9. Determining the number of elements given the number of proper subsets
- 10. Representing information on the Venn diagram
- 11. Solving problems involving Venn diagram
- 12. Probability of simple events
- 13. Word problems on probability
- 14. Finding the probability of events using Venn diagrams

THEME: NUMERACY TOPIC 2: Whole numbers (20 periods)

Subtopics

- 1. Place values and values of digits
- 2. Operation on values of digits
- 3. Reading and writing numbers in words
- 4. Reading and writing words in figures
- 5. Reading and writing Roman numerals
- 6. Converting Hindu Arabic numerals to Roman numerals
- 7. Converting Roman numerals to Hindu Arabic numerals
- 8. Application on Roman numerals
- 9. Revision on basic names of bases
- 10. Place values of digits in other bases
- 11. Expanding numbers in non-decimal base
- 12. Find the numbers that have been expanded in non-decimal base.
- 13. Converting non-decimal base to a decimal base.
- 14. Converting numbers from decimal base to non-decimal bases.
- 15. Converting numbers from non-decimal base to other non-decimal bases.
- 16. Finding unknown bases.
- 17. More on finding unknown bases.
- 18. Addition in non-decimal bases.
- 19. Subtraction in non-decimal bases.
- 20. Multiplication in non-decimal bases.

THEME: NUMERACY TOPIC 3: OPERATION ON WHOLE NUMBERS

Subtopics

- 1. The four basic operations:
 - 1. Addition of whole numbers (with and without regrouping).

- 2. Subtraction of whole numbers (with and without regrouping).
- 3. Multiplication of whole numbers (with and without regrouping).
- 4. Division of whole numbers.
- 2. Mixed operations.
- 3. Properties of whole numbers:
 - 6. Commutative.
 - 7. Associative.
 - 8. Distributive.
- 4. Expanding numbers using indices.
- 5. Writing expanded numbers in short.
- 6. Simplifying expressions involving indices.
- 7. Applying laws of indices.
- 8. Application of indices.
- 9. Writing numbers in standard form (scientific notation).
- 10. Finding numbers that have been written in standard form.
- 11. Prime factorisation.
- 12. Application of prime factors.
- 13. Finding the second number when given LCM, GCF and the first number.

TOPIC: SQUARE ROOTS (CONTINUED)

- 19. Applying LCM to solve daily life problems.
- 20. Square roots of numbers.
- 21. Square roots of fractions.
- 22. Square roots of decimals.
- 23. Solving problems involving square roots.

THEME: NUMERACY

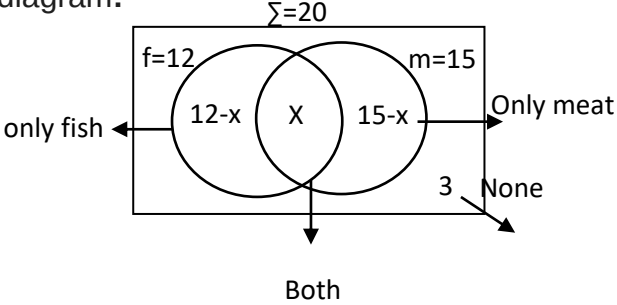
TOPIC 4: NUMBER PATTERNS AND SEQUENCES (11 periods)

- 1. Divisibility tests (for 6, 8, 9, 10 and 11).
- 2. Types of numbers.
- 3. Finding sum of consecutive numbers using formula.
- 4. Factors and multipliers of numbers.
- 5. **Prime factorisation.**
- 6. **Writing prime factors of numbers in;**
 - o (a) Subscript form / set notation.
 - o (b) Power form.
- 7. **Representing prime factors of numbers on Venn diagrams.**
- 8. **Finding unknown factors from the Venn diagrams, LCM and GCF.**
- 9. **Application of LCM and GCF.**
- 10. Finding consecutive numbers.
- 11. Finding cube roots.

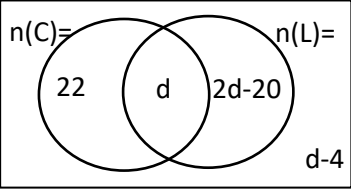
WK	PD	THEME	TOPIC	SUB TOPIC	COMPETENCES SUBJECT	COMPETENCES LANGUAGE	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
1	1	SETS	SET CONCEPTS	FINITE SETS	The Learner; Defines finite sets. - Identifies finite sets. - Forms finite sets on their own.	The Learner; - Reads the words on sets correctly. - sets - infinite - End - Counted - listed	Finite sets Finite sets are sets with exact end. Finite sets are sets whose members can all be listed and counted. Example: 1. A set of days of the week. {Sun, Mon, Tue, Wed, Thur, Fri, Sat}. 7 elements. 2. A set of prime numbers less than 10. {2, 3, 5, 7}. 4 elements.	- Guided discovery - Guided discussion Think, pair, share	- Counting members in a given set. - Drawing pictures of a given set. - Forming a given finite sets.	- Identifying finite sets. - Listing finite sets.	- Charts - Flash cards	Life skills: - Creative thinking Critical thinking Problem solving Values: Patience Responsibility	Founta in Primar y Maths book 7 page 1	
	2		SET CONCEPTS	INFINITE SETS	The Learner; - Defines infinite sets. - Identifies infinite sets.	The Learner; - Reads the words related to sets correctly. - Sets - infinite - Endless - listed - Uncountable	Infinite sets Infinite sets are sets with endless members/elements. Are also sets whose members can't all be listed and counted? Example A set of all counting numbers. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ...} 2. A set of all composite numbers. {4, 6, 8, 9, 10, 12, 14, 15, 16, 18, ...} The members in the above two sets are endless.			- Identifying infinite sets. - Listing infinite sets.		Life skills: - Ability - Cooperation	Primar y Curricu lum page 136-138	

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1	3	SETS	SET CONCEPTS	FORMING AND LISTING SUBSETS	The learner; - Forms subsets from finite sets. - Counts the subsets formed.	The learner; - Describes subsets. - Uses the words in the sentence correctly.	Subsets A subset is a set that can be formed from another set (mother set). A subset is any part of a given set. Symbol: subset eq Example: Given that set $A = \{1, 2, 4\}$, form and list all the subsets in set A. Soln: $\{\}, \{1\}, \{2\}, \{4\}, \{1,2\}, \{1,4\}, \{2,4\}, \{1,2,4\}$	- Guided discovery - Guided discussion - Demonstration	- Counting - Drawing	- Forming subsets - Listing subsets - Counting the subsets	- Charts - Flash cards	Life skills: - Critical thinking - Creative thinking - Problem solving Values: - Patience	MK book 7 page 4; P.7 curriculum page 136-138	
	4			USING THE FORMULA TO FIND THE NUMBER OF SUBSETS	The learner; - Derives the formula for finding the subsets. - Uses the formula to find the number of subsets.	The learner; - Explains the steps taken to find the number of subsets. - Derives formula. - Subset. - Element.	Examples 1. A set has 6 elements. Find the number of subsets in that set. Soln: No. of subsets $= 2^n = 2^6$ $= 2 \times 2 \times 2 \times 2 \times 2 \times 2$ $= 4 \times 4 \times 4$ $= 16 \times 4$ $= 64$ subsets 2. Given that set $K = \{p, q, r, s\}$. Find the number of subsets in set K. Soln: No. of subsets $= 2^n = 2^4$ $= 2 \times 2 \times 2 \times 2$ $= 4 \times 4$ $= 16$	- Guided discovery - Guided discussion - Demonstration	- Drawing - Counting	- Deriving the formula for finding subsets. - Using the formula to find the number of subsets.		Life skills: - Responsibility - Co-operation	MK book 7 page 5; P.7 curriculum page 136-138	

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1	5	SETS	SET CONCEPTS	Finding the number of elements given the number of subsets	The learner - applies the formula to find the number of elements given the number of subsets	The learner; - Explains the steps taken to find the number of elements.	Examples 1. Given that set P has 16 subsets. Find the number of elements in set P. Solution: No. of subsets = 2^n $16 = 2^n$ (Prime factorize 16: $2 \times 2 \times 2 \times 2 = 2^4$) $2^4 = 2^n$ n = 4 Set P has 4 elements.	- Guided discovery - Guided discussion - Explanation	- Drawing - Counting of elements when given the subsets	- Finding the number of elements when given the subsets	- Charts - Flash cards	Life skills: - Creative thinking - Critical thinking Values: - Problem solving	MK book 7 page 7; P.7 curriculum page 136-138	
	6		SET CONCEPTS	Forming and listing proper subsets	The learner; - Forms proper subsets from the given sets.	The learner; - Describes proper subsets. - Uses the words related to sets correctly.	Proper Subsets A proper subset is a set with fewer/less members than the given set. Symbol: subset Example: 1. Given that set M = {a, b, c}. Form and list the proper subsets in set M. Soln: { }, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}	- Guided discovery - Guided discussion - Explanation		- Forming and listing the proper subsets	- Text books	Creative thinking	MK book 7 page 5; P.7 curriculum page 136-138	
	7		SET CONCEPTS		The learner; - Derives the formula for finding the number of proper subsets. - Uses the formula to find the number of proper subsets.	The learner; - Uses the words correctly. - Pronounces the words correctly.	Examples 1. Given that Q = {2, 3, 5, 7}. Find the number of proper subsets in set Q. Solution: No. of proper subsets = $2^n - 1$ = $2^4 - 1$ = $(2 \times 2 \times 2 \times 2) - 1$ = $(4 \times 4) - 1$ = $16 - 1$ = 15 proper subsets	- Guided discovery - Guided discussion		- Applying the formula to find the number of proper subsets.			MK book 7 page 6	

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2	1	SETS	SET CONCEPTS	Finding the number of elements given proper subsets	The learner; - applies the formula for finding number of elements given proper subsets.	The learner; - Explains the steps for finding the number of elements given proper subsets.	Example 1. Given that set B has 7 proper subsets. How many elements are in set B? Solution: $\text{No. of proper subsets} = 2^n - 1$ $7 = 2^n - 1$ $7 + 1 = 2^n$ $8 = 2^n$ (Prime factorize 8: $2 \times 2 \times 2 = 2^3$) $2^3 = 2^n$ n = 3 Therefore Set B has 3 elements.	- Guided discovery - Guided discussion - Explanation	- Drawing - Counting	- Finding the number of elements - Finding the number of proper subsets	- Charts - Flash cards	Life skills: - Critical thinking - Creative thinking Values: - Problem solving	MK book 7 page 7	
	2			Representing information on a Venn diagram	The learner; - Represents information on a Venn diagram. - Describes parts of the Venn diagram.	The learner; - Reads sentences related to Venn diagrams correctly. - M only - F only	Example In a class of 20 pupils, 12 eat fish (F), 15 eat meat (M), x eat both and 3 eat none of them. a) Represent this information on a Venn diagram:  b) Find the value of x: $12 - x + x + 15 - x + 3 = 20$ $(12 + 15 + 3) - x = 20$ $30 - x = 20$ $30 - 30 - x = 20 - 30$ $-x = -10$ x = 10	- Guided discovery - Guided discussion - Explanation		- Representing the number given information on a Venn diagram		Life skills: - Creative thinking - Critical thinking Values: - Problem solving	MK book 7 page 8	

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2	3	SETS	SET CONCEPTS	Solving problems involving Venn diagrams	The learner; - Solves problems involving Venn diagrams correctly.	The learner; - Reads sentences related to Venn diagrams correctly.	Example 1. Use the Venn diagram below to answer the questions that follow: a) Find the value of h: $h + 2h + 1 + 8 + h + 2 = 16$ $3h + 1 + 10 = 16$ $3h + 11 = 16$ $3h + 11 - 11 = 16 - 11$ $3h = 5$ $h = 5/3$ h = 5 b) Find n(N): $8 + h = 8 + 5 = 13$ c) Find the value of k: $n(\Sigma) = K$ $= 16 + h + 8 + 2h + 2$ $= 16 + 5 + 8 + 10 + 2$ $= 31$	- Guided discovery - Guided discussion - Explanation - Demonstration	- Drawing - Solving real life problems involving Venn diagrams	- Solving Venn diagrams.	- Charts	Life skills: - Critical thinking - Creative thinking Values: - Problem solving	MK book 7 page 11	
2	4 & 5	SETS	SET CONCEPTS	Probability of simple events	The learner; - Works out probability of simple events.	The learner; - Reads and uses the words related to probability correctly.	Example 1. The probability of choosing a girl from class to be a class prefect is 17/30. What is the probability of choosing a boy from the class? Solution: Probability of a boy = $1 - 17/30$ $= 30/30 - 17/30 = 13/30$ 2. A dice is tossed once. Find the probability that a prime number will appear on top. S.S = {1, 2, 3, 4, 5, 6} Events = {2, 3, 5} Prob = $n(\text{Events})/n(\text{S.S}) = 3/6$	- Guided discovery - Guided discussion - Explanation		- Calculating probability of numbers.	- Coins - Dice	Life skills: - Critical thinking - Creative thinking	MK book 7 page 10, 154-156	

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2	6 & 7	SETS	SET CONCEPTS	Working out probability of events using Venn diagrams	The learner; - works out probability using Venn diagrams correctly.	The learner; - Reads sentences related to Venn diagrams and probability correctly. - Tells the probability of simple events.	Example 1. In a bus, 22 passengers bought chicken (C) only, d passengers bought both chicken and liver (L), (2d - 20) bought liver only while d - 4 bought neither of the two. a) Complete the Venn diagram below:  b) If the number of passengers who bought liver is twice the passengers who did not buy any of the two types of food, find the value of d. Soln $d + 2d - 20 = 2(d - 4)$ $3d - 20 = 2d - 8$ $3d - 2d - 20 = 2d - 2d - 8$ $d - 20 = -8$ $d - 20 + 20 = -8 + 20$ d = 12 c) Find the probability that a passenger picked at random from the bus bought chicken. $n(\Sigma) = 22 + d + 2d - 20 + d - 4$ $= 22 + 12 + (2 \times 12) - 20 + 12 - 4$ $= 34 + 24 - 20 + 8 = 46$ $n(\Sigma) = 22 + d = 22 + 12 = 34$ $\text{Prob} = n(\text{events})/n(\text{s.s}) = \frac{34}{46}$	- Guided discovery - Guided discussion - Explanation - Demonstration	- Drawing - Counting	- Working out probability of events using Venn diagrams. - Solving Venn diagrams.	- Charts	Life skills: - Critical thinking - Creative thinking Values: - Problem solving	MK book 7 page 11; P.7 curriculum page 136-138	

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3	1	NUMERACY	WHOLE NUMBERS	Place values and values of digits	The learner; - Identifies place values of the given digits correctly. - Finds values of the given digits.	The learner; - Reads the words correctly.	Examples 1. What is the place value of each digit in 34,561,708 ? <table border="1"><tr><th colspan="3">Millions</th><th colspan="3">Thousands</th><th colspan="3">Units</th></tr><tr><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td>3</td><td>4</td><td>5</td><td>6</td><td>1</td><td>7</td><td>0</td><td>8</td></tr></table> 3 - Ten millions 4 - Millions 5 - Hundred thousands 6 - Ten thousands 1 - Thousands 7 - Hundreds 0 - Tens 8 - Ones 2. Find the value of each digit in 4,876,213 . Solution: <table border="1"><tr><th colspan="3">Millions</th><th colspan="3">Thousands</th><th colspan="3">Units</th></tr><tr><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td></td><td>4</td><td>8</td><td>7</td><td>6</td><td>2</td><td>1</td><td>3</td></tr></table> 3 x 1 = 3 1 x 10 = 10 2 x 100 = 200 6 x 1000 = 6,000 7 x 10,000 = 70,000 8 x 100,000 = 800,000 4 x 1,000,000 = 4,000,000	Millions			Thousands			Units			H	T	O	H	T	O	H	T	O		3	4	5	6	1	7	0	8	Millions			Thousands			Units			H	T	O	H	T	O	H	T	O			4	8	7	6	2	1	3	- Guided discovery - Guided discussion - Explanation	- Drawing - Counting	- Writing place values of digits. - Finding values of given digits.	- MK bk 7 pg 13-14 - Flash cards - Abacus	Life skills: - Critical thinking - Creative thinking Values: - Effective communication	MK book 7 page 13-14	
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3	2	NUMERACY	WHOLE NUMBERS	Operation on values of digits	The learner; - Works out the values of digits. - Adds the values of digits. - Subtracts the values of digits. - Multiplies the values of digits. - Divides the values of digits.	The learner; - Reads the words and numerals correctly.	Examples 1. Find the sum of the values of 4 and 6 in the number 63,478 . Soln: Value of 4 = 4 x 100 = 400. Value of 6 = 6 x 10,000 = 60,000. Sum = 60,000 + 400 = 60,400. 2. Given that 84,137,509 , work out the product of the value of 4 and the value of 9. <table border="1"><tr><td colspan="3">Millions</td><td colspan="3">Thousand s</td><td colspan="3">Units</td></tr><tr><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td>8</td><td>4</td><td>1</td><td>3</td><td>7</td><td>5</td><td>0</td><td>9</td></tr></table> Soln: Value of 4 = 4 x 1,000,000 = 4,000,000. Value of 9 = 9 x 1 = 9. Product = 4,000,000 x 9 = 36,000,000. EXERCISE: 1. Find the sum of the value of 9 and value of 8 in the number 947,856 . 2. The value of 6 in a number is 6,000. What is its place value? 3. Find the quotient of the value of 3 and 5 in the number 3,205 .	Millions			Thousand s			Units			H	T	O	H	T	O	H	T	O		8	4	1	3	7	5	0	9	- Guided discovery - Guided discussion - Explanation	- Drawing - Counting	- Finding the values of digits. - Adding the values. - Subtracting the values. - Multiplying the values of digits.	- MK bk 7 pg 14 - Flash cards - Counter s	Life skills: - Critical thinking - Creative thinking - Problem solving Values: - Fluency	MK boo k 7 pag e 139-141	
Millions			Thousand s			Units																																			
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3	3	NUMERACY	WHOLE NUMBERS	Reading and writing numbers in words	The learner; - Counts numbers up to 99,999,999 in words. - Writes numbers in words up to 99,999,999.	The learner; - Reads numbers up to 99,999,999 in words correctly.	Examples 1. Write 19,058,832 in words. <table><tr><td colspan="3">Millions</td><td colspan="3">Thousand s</td><td colspan="3">Units</td></tr><tr><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td>1</td><td>9</td><td>0</td><td>5</td><td>8</td><td>8</td><td>3</td><td>2</td></tr></table> 19 - Nineteen million 058 - Fifty-eight thousand 832 - Eight hundred thirty-two Ans: Nineteen million, fifty-eight thousand, eight hundred thirty-two. 2. A firm bought a plot of land at Sh. 25,769,000. Express this in words. <table><tr><td colspan="3">Millions</td><td colspan="3">Thousand s</td><td colspan="3">Units</td></tr><tr><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td>2</td><td>5</td><td>7</td><td>6</td><td>9</td><td>0</td><td>0</td><td>0</td></tr></table> Ans: Twenty-five million, seven hundred sixty-nine thousand shillings.	Millions			Thousand s			Units			H	T	O	H	T	O	H	T	O		1	9	0	5	8	8	3	2	Millions			Thousand s			Units			H	T	O	H	T	O	H	T	O		2	5	7	6	9	0	0	0	- Guided discovery - Guided discussion - Explanation	- Drawing - Counting - Reading numbers in words.	- MK bk 7 pg 15 - Chart s - Flash cards	Life skills: - Critical thinking - Creative thinking Values: - Effective communication	MK book 7 page 15	
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3	4	NUMERACY	WHOLE NUMBERS	Reading and writing numbers in figures	The learner; - Writes number words in figures correctly.	The learner; - Reads the words correctly. - Spells the words correctly.	Example 1. Write forty-one million, five hundred twenty-six thousand, eight hundred thirty-two in figures Soln Forty-one million - 41,000,000 Five hundred twenty-six thousand - 526,000 Eight hundred thirty-two - + 832 Ans: 41,526,832	- Guided discovery - Guided discussion - Demonstration		- Reading and writing number words and in figures.	- MK bk 7 pg 16	Values: - Problem solving	MK book 7 page 16																																																						

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3	4	NUMERACY	WHOLE NUMBERS	Reading and writing numbers in figures	The learner; - Writes number words in figures correctly.	The learner; - Reads the words correctly.	Example 1. Write eighty-four thousand, thirty-six in figures. Solution: $\begin{array}{r} 84,000 \\ + 36 \\ \hline 84,036 \end{array}$	- Guided discovery - Demonstration		- Writing number words in figures.	- MK bk 7 - Flash cards	Values: - Problem solving	MK book 7 page 16	
3	5	NUMERACY	WHOLE NUMBERS	Reading and writing Roman numerals	The learner; - Counts Roman numerals up to MM. - Reads Roman numerals up to MM.	The learner; - Reads the Roman numerals correctly.	Roman numerals: I=1, V=5, X=10, L=50, C=100, D=500, M=1000 Repeated Roman numerals: 2=II, 3=III, 20=XX, 30=XXX, 200=CC, 300=CCC, 2000=MM, 3000=MMM Subtractions: 4 (5-1) = IV 9 (10-1) = IX 40 (50-10) = XL 90 (100-10) = XC 400 (500-100) = CD 900 (1000-100) = CM	- Guided discovery - Guided discussion - Explanation	- Drawing - Counting	- Writing Roman numerals up to MM. - Reading Roman numerals.	- Charts - Flash cards - Counters	Life skills: - Creative thinking - Critical thinking Values: - Endurance - Appreciation	MK book 7 page 17; NCAC bk 7 pg 139-141	

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WK	PD	THEME	TOPIC	SUB TOPIC	COMPETENCES (SUBJECT)	COMPETENCES (LANGUAGE)	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
3	7	NUMERACY	WHOLE NUMBERS	Converting Roman numerals to Hindu-Arabic numerals	The learner; - Converts Roman numerals to Hindu -Arabic numerals correctly.	The learner; - Reads Roman numerals to Hindu -Arabic numerals correctly.	Examples: 1. Express CMLXXV as Hindu-Arabic numerals. = CM L XX V = 900 + 50 + 20 + 5 = 975 2. Angela paid CCCL shillings for a cake. Write the amount of money she paid in Hindu-Arabic numerals. Soln: C + C + C + L = 100 + 100 + 100 + 50 = 350	- Guided discovery - Guided discussion	- Drawing - Counting	- Converting Roman numerals to Hindu-Arabic numerals.	- Charts - Flash cards	Life skills: - Creative thinking - Critical thinking Values: - Problem solving	MK book 7 page 19; Curriculum pg 139-141	
4	1	NUMERACY	WHOLE NUMBERS	Revision on basic names of bases and digits used	The learner; - Writes the basic names of bases correctly. - Writes the digits used in each base correctly.	The learner; - Reads and writes the basic names of bases correctly.	Revision on basic names of bases and digits used in each base: Base two - Binary base {0, 1} Base three - Ternary base {0, 1, 2} Base four - Quaternary base {0, 1, 2, 3} Base five - Quinary base {0, 1, 2, 3, 4} Base six Senary base {0, 1, 2, 3, 4, 5} Base seven - Septenary base {0, 1, 2, 3, 4, 5, 6} Base eight - Octal base {0, 1, 2, 3, 4, 5, 6, 7} Base nine - Nonary base {0, 1, 2, 3, 4, 5, 6, 7, 8} Base ten - Denary (decimal) base {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}	- Guided discovery - Guided discussion - Explanation		- Writing basic names of bases. - Writing digits used in each base.	- Charts	Life skills: - Creative thinking - Critical thinking	MK book 7; Curriculum pg 139-141	

WK	P D	THEME	TOPIC	SUB TOPIC	COMPETENCES (SUBJECT)	COMPETENCES (LANGUAGE)	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
4	2	NUMERACY	WHOLE NUMBERS	Place values of digits in non-decimal bases	The learner; - identifies the place value of each digit in non-decimal bases correctly.	The learner; - Reads the place values in non-decimal bases correctly. - reads numbers in non-decimal bases in words.	Examples: 1. Write the place value of each digit in: 1101_{two} 1 - Ones (2^0) 0 - Twos (2^1) 1 – Two-twos (2^2) 1 – Two two - twos (2^3) 2. Write the place value of each digit in: 3205_{seven} 5 - ones (7^0) 0 - sevens (7^1) 2 - Forty-nines (7^2) 3 - Three hundred forty-threes (7^3)	- Guided discovery - Guided discussion	- Drawing - Counting	- Finding place values of digits in non-decimal bases.	- Charts - Flash cards	Life skills: - Creative thinking - Problem solving Values: - Appreciation	MK primary maths bk 7 pg 21	
4	3	NUMERACY	WHOLE NUMBERS	Expanding numbers in non-decimal bases	The learner; - Expands non-decimal base numbers using exponents correctly.	The learner; - Reads the non-decimal base numbers correctly.	Examples: 1. Expand 234_{two} using exponents. Soln: Place values: $4(5^0)$, $3(5^1)$, $2(5^2)$ Ans: $(2 \times 5^2) + (3 \times 5^1) + (4 \times 5^0)$ 2. Expand 1011_{two} using values of the given base. Soln: $1(2^3) + 0(2^2) + 1(2^1) + 1(2^0)$ $= (1 \times 8) + (0 \times 4) + (1 \times 2) + (1 \times 1)$ Ans: $8 + 0 + 2 + 1$	- Guided discovery - Guided discussion - Explanation		- Expanding numbers in non-decimal bases.	- Charts	Life skills: - Creative thinking - Endurance Values: - Appreciation	MK bk 7; Curriculum pg 139-141	

WK	PD	THEME	TOPIC	SUB TOPIC	COMPETENCE S (SUBJECT)	COMPETENCE S (LANGUAGE)	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
4	5	NUMERACY	WHOLE NUMBERS	Converting non-decimal base to a decimal base	The learner; - Converts non-decimal base numbers to decimal base correctly.	The learner; - Reads the non-decimal and decimal base numbers correctly.	Examples: 1. Change 1011_{two} to base ten. Soln: $(1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$ $= (1 \times 8) + (0 \times 4) + (1 \times 2) + (1 \times 1)$ $= 8 + 0 + 2 + 1$ $= 11_{\text{ten}}$ 2. Convert 124_{two} to decimal base. Soln: $(1 \times 5^2) + (2 \times 5^1) + (4 \times 5^0)$ $= (1 \times 25) + (2 \times 5) + (4 \times 1)$ $25 + 10 + 4 = 39_{\text{ten}}$	- Guided discovery - Guided discussion	- Drawing - Counting	- Converting non-decimal bases to base ten. - Charts - Flash cards	Life skills: - Creative thinking - Problem solving Values: - Appreciation	MK bk 7; Uganda Primary Maths bk 7 pg 24		
4	6	NUMERACY	WHOLE NUMBERS	Converting decimal base to non-decimal base	The learner; - Converts decimal numbers to non-decimal base correctly.	The learner; - Reads the numbers in decimal and non-decimal bases correctly.	Examples: Convert 17_{ten} to binary base.	Base	Number	Rem		:---	:---	:---

WK	PD	THEME	TOPIC	SUB TOPIC	COMPETENCES (SUBJECT)	COMPETENCES (LANGUAGE)	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	RE M															
4	7	NUMERACY	WHOLE NUMBERS	Changing non-decimal bases to non-decimal bases	The learner; - Converts non-decimal bases to other non-decimal bases correctly.	The learner; - Explains the steps taken while converting bases.	Note: To change from one non-decimal base to another, first change to base ten, then to the required base. Examples: 1. Change 34_{two} to binary base. Step 1: Change to base ten $(3 \times 5^1) + (4 \times 5^0)$ $= 15 + 4$ $= 19_{\text{ten}}$ Step 2: Change 19_{ten} to base two <table><tr><td>2</td><td>19</td><td>rem</td></tr><tr><td>2</td><td>9</td><td>1</td></tr><tr><td>2</td><td>4</td><td>1</td></tr><tr><td>2</td><td>2</td><td>0</td></tr><tr><td></td><td>1</td><td>0</td></tr></table> $= 10011_{\text{two}}$	2	19	rem	2	9	1	2	4	1	2	2	0		1	0	- Guided discovery - Small group guided discussion - Think, pair, share	Discussing the steps taken while converting	Changing non-decimal bases to non-decimal bases.	Counter	Skills Effective communication Appreciation Creative thinking Values Accuracy Fluency Empathy.		
2	19	rem																											
2	9	1																											
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WK	P D	THEME	TOP IC	SUB TOPIC	COMPETENCE S (SUBJECT)	COMPETENC ES (LANGUAGE)	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
5	1	NUMERACY	WHOLE NUMBERS	Finding unknown bases	The learner; - Finds the unknown bases correctly.	The learner; - Explains the steps taken to find the unknown bases.	Examples: 1. If $21_n = 13_{10}$, find base n. Soln: $(2 \times n^1) + (1 \times n^0) = 13$ $2n + 1 = 13$ $2n + 1 - 1 = 13 - 1$ $2n/2 = 12/2$ n = 6 Therefore base is base six. 2. Given that $32_y = 23_{\text{ten}}$, find base y. Soln: $(3 \times y^1) + (2 \times y^0) = 23$ $3y + 2 = 23$ $3y + 2 - 2 = 23 - 2$ $3y/3 = 21/3$ y = 7 Therefore y is base seven. Example 3: Find base p if $203_p = 35_{\text{ten}}$.	- Guided discovery - Guided discussion - Explanation - Question and answer		- Finding unknown bases.	- Charts	Life skills: - Effective communication - Problem solving Values: - Appreciation	MK bk 7; Curriculum pg 139-141	

WK	P D	THEME	TOPIC	SUB TOPIC	COMPETENCES (SUBJECT)	COMPETENCES (LANGUAGE)	CONTENT	METHO DS AND TECHNI QUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IM S	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
5	2	NUMERACY	WHOLE NUMBERS	More about finding unknown bases	The learner; - Finds the unknown base involving square roots correctly.	The learner; - Explains the steps taken while finding the unknown base.	Examples: 1. Given that $203_p = 35_{\text{ten}}$, find base p. Soln: $(2 \times p^2) + (0 \times p^1) + (3 \times p^0) = 35$ $2p^2 + 0 + 3 = 35$ $2p^2 + 3 = 35$ $2p^2 + 3 - 3 = 35 - 3$ $\frac{2p^2}{2} = \frac{32}{2}$ $\sqrt{p^2} = \sqrt{16}$ p = 4 Therefore base is base four. 2. Work out the value of m given that $2m1_{\text{four}} = 45_{\text{ten}}$. Soln: $(2 \times 4^2) + (m \times 4^1) + (1 \times 4^0) = 45$ $(2 \times 16) + 4m + 1 = 45$ $32 + 4m + 1 = 45$ $33 + 4m = 45$ $33 - 33 + 4m = 45 - 33$ $4m / 4 = 12 / 4$ m = 3	- Guided discovery - Demonstration - Explanation	- Drawing	- Finding unknown bases involving square roots. - Counting.	- Charts	Life skills: - Critical thinking - Effective communication Values: - Appreciation - Problem solving	MK bk 7; Curriculum pg 139-141	

WK	PD	THEME	TOPIC	SUB TOPIC	COMPETENCES (SUBJECT)	COMPETENCES (LANGUAGE)	CONTENT	METHODS AND TECHNIQUES	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM
5	3	NUMERACY	WHOLE NUMBERS	Addition of bases	The learner; - Adds non-decimal bases correctly.	The learner; - Reads and spells mathematical words correctly.	Example 1: Add $1011_{\text{two}} + 1111_{\text{two}}$ Working: $\begin{array}{r} 1011_{\text{two}} \\ + 1111_{\text{two}} \\ \hline 11010_{\text{two}} \end{array}$ $1+1 = 2 \text{ div } 2 = 1 \text{ text\{ rem \} } 0$ (write 0 carry 1) $1+1+1 = 3 \text{ div } 2 = 1 \text{ text\{ rem \} } 1$ (write 1 carry 1) $1+0+1 = 2 \text{ div } 2 = 1 \text{ text\{ rem \} } 0$ (write 0 carry 1) $1+1+1 = 3 \text{ div } 2 = 1 \text{ text\{ rem \} } 1$ (write 1 carry 1) Ans: 11010_{two}	* Guided discovery * Guided discussion * Demonstration	* Drawing * Adding bases	* Counting * Adding bases in different systems * Filling base tables	* Charts * Flash cards	Life skills: * Creative thinking Values: * Effective communication	* MK Bk 7 pg 22-23 * Fountain Primary Maths Bk 7 pg 35-36	* Guided discovery * Guided discussion * Demonstration
5	4	NUMERACY	WHOLE NUMBERS	Subtraction of bases	The learner; - Subtracts bases correctly.	The learner; - Reads words correctly and uses the term "borrowing"	Example 1: $1101_{\text{two}} - 110_{\text{two}}$ Soln: $\begin{array}{r} 1101_{\text{two}} \\ - 110_{\text{two}} \\ \hline 0111_{\text{two}} \end{array}$ Example 2: Subtract 141_{two} from 200_{two}	- Guided discovery - Explanation - Demonstration	- Subtracting bases	- Subtracting bases - Solving problems involving subtraction of non-decimal bases	- Charts - Flash cards - Counters	Life skills: - Critical thinking - Problem solving Values: - Endurance - Appreciation	MK Bk 7 pg 25; Fountain Bk 7 pg 37-38	

WK	PD	THEME	TOPIC	SUB TOPIC	COMPETENCES (SUBJECT)	COMPETENCE S (LANGUAGE)	CONTENT	METHODS AND TECHNIQUE S	ACTIVITIES (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	REF	REM																
5	5 and 6	NUMERACY	WHOLE NUMBERS	Multiplication of bases	The learner: Multiplies non-decimal bases correctly	The learner: Reads, spells and uses the words related to multiplication of bases correctly	Multiplication of bases Examples I. Multiply $10_{\text{two}} \times 11_{\text{two}}$ II. Work out: $134_{\text{two}} \times 13_{\text{two}}$ III The table below shows multiplication in base four. Complete it correctly. <table><tr><td>X</td><td>1</td><td>2</td><td>3</td></tr><tr><td>1</td><td>1</td><td>2</td><td>-</td></tr><tr><td>2</td><td>2</td><td>-</td><td>-</td></tr><tr><td>3</td><td>3</td><td>-</td><td>-</td></tr></table>	X	1	2	3	1	1	2	-	2	2	-	-	3	3	-	-	x	1	2	3			
X	1	2	3																											
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3	3	-	-																											

W K	P D	THEME	TOPIC	SUB TOPIC	COMPETENCE S (SUBJECT)	COMPETENC ES (LANGUAGE)	CONTENT	METHO DS AND TECHNI QUES	ACTIVITIE S (SPECIAL NEEDS)	ACTIVITIES (SUGGESTED)	IMS	INDICATORS OF LIFE SKILLS AND VALUES	RE F	RE M
6	2	NUMERACY	OPERATION ON WHOLE NUMBERS	Multiplication of whole numbers	The learner: Multiplies numbers correctly.	- Reads the words correctly. - Uses the words correctly.	Multiplication of whole numbers Example 1. Workout: $\begin{array}{r} 3452 \\ \times 275 \\ \hline \end{array}$ 2. A trader spent shs 245,614 on a sack of rice. If he paid for 214 sacks, how much did he spend?	* Guided discovery * Guided discussion * Explanation * Demonstration * Question and answer	* Counting using numbers * Drawing	* Multiplying numbers * Counting * Regrouping	* Flash cards * Sticks * Stones * Bottles	Life Skills: * Problem solving * Effective communication * Critical thinking Values: * Co-operation * Appreciation * Responsibility	MK BK 7	Pg 32

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6	3	NUMERACY	OPERATION ON WHOLE NUMBERS	Division of whole numbers	The learner: Divides whole numbers correctly.	The learner: Reads, pronounces and uses the words related to division correctly.	Example 1: Divide 6363 by 9. Example 2: On a farm, 14167 litres of milk are produced in a month of 31 days. How many litres are produced daily?	Demonstration, Guided discovery, Guided discussion, Explanation, Question and answer.	Counting, Drawing.	Counting, dividing whole numbers correctly.	Flash cards, Stones, Sticks.	Life skills: Creative thinking, problem solving, saving, Effective communication. Values: appreciation, Co-operation, Responsibility.	MK BK 7 Pg 33	

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6	4	NUMERACY	OPERATION ON WHOLE NUMBERS	Mixed operation on whole numbers	The learner: Works out numbers involving mixed operations correctly.	The learner: Reads and explains the steps taken while working out mixed operations.	Examples 1. Work out: $48 - 20 + 22$ Solution: $(48 + 22) - 20$ $70 - 20$ = 50 2. Simplify: $1/2$ of $50 + 39 / 3$ Solution: BODMAS $(1/2 \text{ of } 50) + 39 / 3$ $(1/2 \times 50) + (39 / 3)$ $25 + 13 = \mathbf{38}$	* Guided discovery * Guided discussion * Demonstration * Explanation	* Drawing * Counting	Working out number involving mixed operations.	* Flash cards * Sticks * Stones	Life Skills * Problem solving * Effective communication * Creative thinking	Curriculum Bk 7 Page 142 - 144	
	5	NUMERACY	OPERATION ON WHOLE NUMBERS	Properties of numbers	The learner: Applies the basic operations integrated with commutative and associative properties.	The learner: Describes commutative and associative properties.	Commutative property and associative property. * The order in which two numbers are added or multiplied does not change the sum or product (Commutative property) * When carrying out addition or multiplication of more than two numbers, the way in which the numbers are grouped does not affect the sum or product. Examples 1. $8 + 4 = 4 + 8$ (Commutative property) 2. $4 \times 6 = 6 \times 4$ (Commutative property) 3. $(6 + 4) + 5 = 6 + (4 + 5)$ (Associative property) 4. $(3 \times 4) \times 5 = 3 \times (4 \times 5)$ (Associative property)			Working out operations on numbers integrating with commutative and associative properties.		Values * Appreciation * Cooperation	Curriculum Bk 7 Pg 142 - 144 MK Book 7 Pg 34 - 35	

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6	6	NUMERACY	OPERATIONS ON WHOLE NUMBERS	Distributive Property	The learner: Applies the basic operations integrated with distributive property.	The learner: Describes distributive property. Explains steps taken when working out numbers using distributive property.	In distributive property, the product of the common factor and the sum or difference of any operation is equal to the sum or difference of the product respectively. Examples: i. Work out: 13 X 8 - 3 X 8 Solution: (13 - 3) X 8 = 10 X 8 = 80 ii. Work out: (83 div 5) + (17 div 5) Solution: (83 + 17) div 5 = 100 div 5 = 20	* Guided discovery * Guided discussion * Demonstration * Explanation	* Drawing * Counting 103	Working out numbers integrating with distributive property. 102	Life Skills: * Problem solving * Effective communication * Creative thinking	* Flash cards * Sticks * Stones	Curriculum Bk 7 Page 142 - 144 MK Bk 7 Pg 35																			
	7	NUMERACY		OPERATIONS ON WHOLE NUMBERS	Expanding numbers using exponents	The learner: Writes numbers in expanded form.	The learner: Reads numbers in expanded form. <table border="1"><tr><td>10⁴</td><td>10³</td><td>10²</td><td>10¹</td><td>10⁰</td></tr><tr><td>4</td><td>9</td><td>6</td><td>4</td><td>7</td></tr></table> (4X10 ⁴)+(9X10 ³)X(6X10 ²) +(4X10 ¹)+(7X10 ⁰) Expand 352.48 Using exponents <table border="1"><tr><td>10²</td><td>10¹</td><td>10⁰</td><td>10⁻¹</td><td>10⁻²</td></tr><tr><td>3</td><td>5</td><td>2</td><td>.4</td><td>8</td></tr></table> (3X10 ²)+(5X10 ¹)X(2X10 ⁰) +(4X10 ⁻¹)+(8X10 ⁻²)	10 ⁴	10 ³	10 ²	10 ¹	10 ⁰	4	9	6	4	7	10 ²	10 ¹	10 ⁰	10 ⁻¹	10 ⁻²	3	5	2	.4	8					
10 ⁴	10 ³	10 ²	10 ¹	10 ⁰																												
4	9	6	4	7																												
10 ²	10 ¹	10 ⁰	10 ⁻¹	10 ⁻²																												
3	5	2	.4	8																												

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7	1		NUMERACY	OPERATIONS ON WHOLE NUMBERS	Writing expanded numbers in short	The learner writes expanded numbers in short.	The learner reads expanded numbers in short. Examples: 1. What number is expanded to give $(2 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (4 \times 10^0)$ $20000 + 7000 + 500 + 80 + 4 = 27584$ 2. Express $(9 \times 10^2) + (7 \times 10^0) + (6 \times 10^{-2})$ as a single number. $900 + 7 + 0.06 = 907.06$	* Guided discovery * Guided discussion * Explanation	* Drawing * Counting	Writing expanded numbers in short.	* Flash cards * Sticks * Stones	Life Skills: * Problem solving * Effective communication * Creative thinking	Curriculum bk7 Pg 142-144 MK bk7 Pg 38	
	2		NUMERACY	OPERATIONS ON WHOLE NUMBERS	Simplifying expressions involving indices	The learner simplifies expressions involving indices correctly.	The learner reads indices words/expressions correctly. Examples: 1. Simplify: $4^2 \times 4^3 = 4 \times 4 \times 4 \times 4 \times 4 = 16 \times 16 \times 4 = 1024$ OR: $4^{(2+3)} = 4^5$ 2. Simplify: $\frac{k^3 \times k^4}{k^2}$ $= \frac{k \times k \times k \times k \times k \times k \times k}{k \times k}$ $= k^5$			Simplifying expressions involving indices.		Values: * Appreciation * Cooperation	MK 2000 Page	

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7	3&4	NUMERACY	OPERATION ON WHOLE NUMBERS	Application of laws of indices	The learner: solves equations involving indices.	The learner: explains the steps taken to solve equations involving indices.	Examples: 1. Solve for n; If $2^n = 64$ $\begin{array}{r l} 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$ Soln: $2^n = 2 \times 2 \times 2 \times 2 \times 2 \times 2$ $2^n = 2^6$ $n=6$ 2. Solve for a: $2^a - 1 = 7$ Soln: $2^a = 7 + 1$ $\begin{array}{r l} 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$ $2^a = 8$ $2^a = 2^3$ $a=3$ 3. Find r, if $5^r / 5^3 = 25$ $5^{(r-3)} = 5^2$ $r-3 = 2$ $r = 2+3$ $r=5$	Guided discussion, Discovery, Explanation, Demo.	Drawing, Counting. Solving equations involving indices.	Life skills: Problem solving, effective communication, creative thinking. Values: Appreciation, Cooperation, Responsibility.	Charts, flash cards, sticks.	MK book 7

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5	NUMERACY	OPERATION ON WHOLE NUMBERS	Writing numbers in standard form (Scientific notation)	The learner writes numbers in standard form.	The learner reads numbers in standard form.	Examples: 1. Write 1489 in scientific notation. Soln: $1489 \div 10 = 148.9$ dots math $\{1.489 \times 10^3\}$. 2. Write 0.0678 in standard form. Soln: $0.0678 \times 10 = 0.678$ dots $\{6.78 \times 10^{-2}\}$	Guided discussion, Guided discovery, Explanation.	Drawing, Writing numbers in standard form.	Life Skills: Problem solving, Effective communication, Creative thinking.	Flash cards, sticks, stones.	Curriculum bk7 Pg 142-144, MK bk7 Page 39.
6	NUMERACY	OPERATION ON WHOLE NUMBERS	Finding numbers that have been written in standard form	The learner writes words and numbers written in standard form correctly.	The learner reads the words and numbers written in standard form correctly.	Examples: 1. Express 3.08×10^3 as a single number. Soln: $\frac{308}{100} \times 10 \times 10 \times 10 = 308 \times 10 = 3080$. 2. What number has been expressed in standard form to give 2.3×10^{-4}	Demonstration.	Writing numbers written in standard form in short.			MK bk7 Page.

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W K	P D	THEME	TOPIC	SUB TOPIC	COMPETE NCE (SUBJECT)	COMPETEN CE (LANG)	CONTENT	METHODS / TECHNIQU ES	ACTIVITIES (SPECIAL)	ACTIVITIE S (SUGGES TED)	L. SKILLS & VALUES	IMS	REF	RE M																					
8	2	NUMERACY	OPERATION ON WHOLE NUMBERS	Finding the missing number given L.C.M and G.C.F	The learner; finds the other numbers when given L.C.M, G.C.F and one number.	The learner; Explains the relationship between L.C.M, G.C.F and the numbers.	Examples 1. The L.C.M of two numbers is 72, their G.C.F is 6. If one of the numbers is 24, find the second number. Solution 1 st no x 2 nd no = L.C.M x G.C.F= $\frac{LCM \times GCF}{Given\ no.}$ $\frac{72 \times 6}{24}$ 2no. = 18	Guided discovery Guided discussion	Drawing	Finding the other numbers given the L.C.M and G.C.F	Problem solving Effective communication	Flash cards	P.7 Curriculum Page 142 - 144																						
	3			Application of L.C.M in the daily life	The learner; Applies the L.C.M in daily life. Works out the given problems.	The learner; Reads the given mathematical statements correctly.	Examples Two bells ring at intervals of 30 minutes and 40 minutes respectively for lower and upper primary. After every how long will the bells ring together at the same time? Solution <table><tr><td>2</td><td>30</td><td>40</td></tr><tr><td>2</td><td>15</td><td>20</td></tr><tr><td>2</td><td>15</td><td>10</td></tr><tr><td>2</td><td>15</td><td>5</td></tr><tr><td>3</td><td>15</td><td>5</td></tr><tr><td>5</td><td>5</td><td>5</td></tr><tr><td></td><td>1</td><td>1</td></tr></table>	2	30	40	2	15	20	2	15	10	2	15	5	3	15	5	5	5	5		1	1	Explanation Sharing Demonstration	Counting Sticks	Counting Applying the L.C.M in daily life.	Creative thinking Appreciation Working out the given problems Responsibility	Bottle tops		
2	30	40																																	
2	15	20																																	
2	15	10																																	
2	15	5																																	
3	15	5																																	
5	5	5																																	
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W K	P D	THEME	TOPIC	SUB TOPIC	COMPETE NCE (SUBJECT)	COMPETEN CE (LANG)	CONTENT	METHODS / TECHNIQ UES	ACTIVITIE S (SPECIAL)	ACTIVITI ES (SUGGE STED)	L. SKILLS & VALUES	IMS	REF	RE M
8	4	NUMERACY	OPERATION ON WHOLE NUMBERS	Square roots of whole numbers	The learner; finds the square roots of whole numbers	The learner; Explains the relationships between square roots and square numbers	Examples 1. Find the square root of 196 Solution $\sqrt{196}$ $\sqrt{(2 \times 2) \times (7 \times 7)}$ $\sqrt{2 \times 7}$ 2×7 =14	Demonstration Guided discovery	Grouping Counting	Finding the square roots of numbers	Effective communication Problem solving	Flash cards	MK Book7 page 41	
	5			Square of fractions.	The learner; Finds the square roots of fractions	The learner; Reads the given fractions correctly	1. Find the square root of $2\frac{7}{9}$ Solution $\sqrt{27/9}$ $\sqrt{25/9}$ $\sqrt{5 \times 5/3 \times 3}$ $\sqrt{5/3}$ $5/3$ =12/3	Guided discussion Observation	Drawing	Finding the square roots of fraction	Creative thinking Appreciation	Bottle tops	P.7 Curriculum page142-144	
	6			Square roots of decimals	The learner; Finds the square roots of decimals	The learner; Reads the given decimals correctly	1. Find the square root of 1.44 Solution $\sqrt{1.44}$ $\sqrt{144/100}$ $(2 \times 2) \times (2 \times 2) \times (3 \times 3)$ $(2 \times 2) \times (5 \times 5)$ 2×3 $2 \times 2 \times 3$ 2×5 12 10 =1.2	Explanation		Finding the square roots of decimals	Co-operation		MK Book 7 page 41	

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8	7	NUMER ACY	OPERATION ON WHOLE NUMBERS	solving problems involving square roots	The learner: - Solves problems involving square roots	The learner - Reads the mathematical statements given correctly	Examples 1. A square garden has an area of 64m^2 . Find the length of each side of the garden.	Demonstration Think Pair Share	Drawing	Solving problems involving square roots Grouping	A chart showin g square roots	Critical thinking Problem solving	P7 Curr pg 142 149	
9	1	NUMER ACY	NUMBER PATTERNS AND SEQUENCES	Divisibility tests	The learner: - Finds out the divisibility tests of 6 and 9.	The learner: - Explains the divisibility tests of 6 and 9. - Reads the given words correctly.	Divisibility tests for 6, 9 1. Test for 6 A number is divisible by six if it is both divisible by 2 and 3. The number has to be even and the sum of the digits has to be a multiple of 3. Examples. Without dividing, determine whether 2346 is divisible by six. - The number is even. - Sum of the digits: $2 + 3 + 4 + 6 = 15$ So, 2346 is divisible by six. 2. Divisibility test of 9 A number is divisible by nine if the sum of the digits is a multiple of 9 3. Test whether 234 is divisible by nine. Sum of digits: $2 + 3 + 4 = 9$	Guided discovery Guided discussion Explanation	Counting	Finding out the divisibility tests of 6 and 9. Grouping Counting Reading the given words correctly	Flash cards A chart showin g divisibil ity tests	Effective communication Problem solving	P7 Curr pg 145 147 MK bk 7 pg 43 and 45	

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9	2	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Divisibility tests	The learner: - Finds out the divisibility tests of 8 and 10	The learner: - Explains divisibility tests of 8 and 10	Divisibility test of 8 and 10 1. A number is divisible by ten if it ends with zero. E.g. 490, 670, 44230 etc. 2. A number is divisible by 8 if the number formed by its last three digits is a multiple of 8 (divisible by 8). Example State whether 277512 is exactly divisible by 8. solution 277[512] 512 is a multiple of 8 so, 277512 is exactly divisible by 8.	Guided discovery Guided discussion	Counting	Finding out the divisibility tests of 8 and 10 - Explaining divisibility tests of 8 and 10 - Drawing	Flash cards Cuttings	Problem solving Critical thinking	Mk Book 7 Page 44	
9	3	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Divisibility tests	The learner: - Finds out the divisibility tests of 11	The learner: - Explains the divisibility tests of 11	Divisibility test for 11 A number is divisible by 11 if the sum of its paired digits from right to left is a multiple of 11. Examples 1. Test whether 36663 is exactly divisible by 11. solution pairs of numbers from right to left: 3(66)(63) 063 66 + 3 ----- 132 132 is a multiple of 11, so, 36663 is exactly divisible by 11.	Explanation Demonstration		Finding out the divisibility tests of 11 Explaining divisibility tests of 11		Creative thinking Critical thinking	Mk book 7 Page 46	

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9	4	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Types of numbers (Square numbers)	The learner: - Defines square numbers - Gives examples of square numbers - Forms patterns and sequences using square numbers	The learner: - Reads patterns and sequences of numbers - Defines square numbers	Square numbers Square numbers are numbers obtained by multiplying a counting number by itself.	Guided discussion Thinking	Giving examples of triangular numbers - Forming patterns and sequences - Defining triangular numbers	Critical thinking Creative thinking	A chart showing square numbers	MK Book 7 Page 47	
	5	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Triangular numbers	The learner: - Gives examples of triangular numbers - Forms patterns and sequences using triangular numbers	The learner: - Defines triangular numbers - Reads patterns and sequences of numbers	Triangular numbers Triangular numbers are numbers obtained by adding consecutive counting numbers. 1, 1+2, 1+2+3, 1+2+3+4, 1+2+3+4+5 1, 3, 6, 10, 15 etc. 1. Find the 9th triangular number. — Solution $\frac{n(n+1)}{2}$ $\frac{9(9+1)}{2}$ $\frac{9 \times 10}{2}$ $\frac{9 \times 5}{1} = 45$ 2. Find the next number in the sequence. 28, 21, 15, 10, 6 (Triangular numbers)	Guided discussion Sharing	Suggested: - Giving examples of triangular numbers - Forming patterns and sequences - Defining triangular numbers	Problem solving Effective communication	A chart showing triangular numbers	MK Book 7 Page 49	

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9	6 and 7	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Cubic numbers and cube roots	The learner; - Gives examples of cubic numbers. - Forms patterns and sequences of numbers using cubic numbers. - Finds cube roots of given numbers.	The learner; - Defines cubic numbers. -Reads patterns and sequences of numbers.	Cubic numbers obtained by multiplying a counting number by itself three x. i.e. $1 \times 1 \times 1$, $2 \times 2 \times 2$, $3 \times 3 \times 3$, $4 \times 4 \times 4$, $5 \times 5 \times 5$ 1, 8, 27, 64, 125 etc. 1. Find the next number the sequence below: 1, 8, 27, 64, 125 (cubic numbers) 2. Find the cube root of 8. solution $3\sqrt{(8)} = 3\sqrt{(2 \times 2 \times 2)}$ $= 3\sqrt{(23)}$ $= 2$ 3. Find the cube root of 125. solution $3\sqrt{(125)}$ $= 3\sqrt{(5 \times 5 \times 5)}$ $= 3\sqrt{(53)}$ $= 5$	Guided discovery Guided discussion Think-Pair-Share Demonstration	Grouping Sharing Drawing	-Giving examples of cubic numbers. -Forming patterns and sequences of numbers using cubic numbers. -Finding cube roots of given numbers. - Defining cubic numbers	Flash cards A chart showing cubic numbers	Critical thinking Creative thinking Problem solving Effective communication	MK Book 7 Page 48	

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10	1	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Prime numbers and composite numbers	The learner; -Gives examples of prime numbers and composite numbers -Forms patterns and sequences using prime numbers and composite numbers	-Defines prime numbers. -Defines composite numbers. -Reads patterns and sequences of numbers.	Prime numbers are numbers with only two factors 1 and the number itself i.e.; 2, 3, 5, 7, 11, 13, 17, 19, 23 etc. 1. Find the next number in the sequence; 13, 15, 18, 23, 30, 41 (differences: +2, +3, +5, +7, +11 (prime no's)) $30 + 11 = 41$ Composite numbers are numbers with more than two factors. i.e.; 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, etc. 1. Find the next number in the sequence. 10, 14, 20, 28, 37, 47, 59 (differences: +4, +6, +8, +9, +10, +12 (composite no's)) $47 + 12 = 59$ 2. Find the sum of the 5th and the 8th composite numbers solution 4, 6, 8, 9, (10), 12, 14, (15), 16, 18 (5th is 10, 8th is 15) Sum: $10 + 15 = 25$	Guided discovery Think-Pair-Share Guided discussion	Sharing Grouping Counting	-Giving examples of prime numbers and composite numbers -Forming patterns and sequences -Defining prime numbers -Defining composite numbers	Critical thinking Creative thinking Problem solving	A chart showing prime numbers and composite numbers Flash cards	MK Bk 7 page 50	

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10	2 and 3	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Whole numbers, counting numbers and consecutive natural numbers	The learner; -Gives examples of whole numbers and counting numbers. -Forms patterns and sequences using whole and counting numbers. -Finds consecutive even numbers when given total	The learner; -Defines whole numbers and counting numbers. -Explains steps taken when finding consecutive numbers.	Whole numbers are numbers that begin from zero (0) and move on consecutively i.e.; 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, etc. Counting numbers are numbers which begin from one (1) and move on consecutively i.e. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 ... 1. The sum of 4 consecutive counting numbers is 126. a) Find the 4 consecutive counting numbers solution Let the 1st consecutive number be P 1st: P, 2nd: P+1, 3rd: P+2, 4th: P+3	Sum = 126 $P+P+1+P+2+P+3 = 126$ $4P + 6 = 126$ $4P + 6 - 6 = 126 - 6$ $\frac{4P}{4} = \frac{120}{4}$ P = 30 Numbers P, P+1, P+2, P+3 30, 31, 32, 33	Guided discovery Guided discussion Explanation	Drawing Counting Grouping	Forming patterns and sequences using even numbers Finding consecutive counting numbers when given total	Problem solving Critical thinking Creative thinking Effective communication	Flash cards Sticks	P. 7 Curr. Pg 145 148

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10	4 and 5	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Even numbers and consecutive even numbers	The learner: Gives examples of even numbers. Forms patterns and sequences using even numbers. Finds consecutive even numbers when given sum.	The learner: Defines even numbers. Explains steps taken when finding consecutive even numbers.	Even numbers are numbers which are exactly divisible by 2. I.e. They end with either: 0, 2, 4, 6 or 8. Example: 0, 2, 4, 6, 8, 10, 16, 32, 44... Qn.1 Find the sum of the first 5 even numbers. solution {0, 2, 4, 6, 8} sum $0 + 2 + 4 + 6 + 8 = 20$ Qn.2 The sum of 3 consecutive even numbers is 42. Find the numbers. Find the range of the numbers	Guided discovery Guided discussion Explanation	Drawing Grouping Counting	Forming patterns and sequences using even numbers Finding consecutive even numbers when given total.	Flash cards Sticks bottles	Problem solving Effective communication Critical thinking Creative thinking	Mk bk7pg 51-53	

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10	6 and 7	NUMERACY	NUMBER PATTERNS AND SEQUENCES	Odd numbers and consecutive odd numbers	The learner: * Gives examples of odd numbers. * Forms patterns and sequences using odd numbers. * Finds consecutive odd № when given sum.	The learner: * Defines odd numbers.	Odd numbers are numbers which are not exactly divisible by 2 i.e., leaves a remainder of 1. i.e., 1, 3, 5, 7, 9, 11, 13, 15, 17... 1. Find the next number in the sequence. 41, 42, 45, 50, 57, 66, 77 (Steps: +1, +3, +5, +7, +9, +11 (odd №)) Qn. Find the three consecutive odd numbers whose sum is 33. Solution Let the 1st consecutive odd number be y.	Guided discovery Guided discussion Explanation	Drawing Grouping Counting Forming patterns and sequences using even numbers Finding consecutive even numbers when given total.	Problem solving Effective communication Critical thinking Creative thinking	Flash cards Sticks bottles	Mk bk7pg 51-53	